Data and Computer Communications

Data Encoding

Signaling revisited

Analog signal: Represents data with continuously varying EM wave

- Analog data ☐ Telephone ☐Analog signal
- Digital data ☐Modem ☐ Analog signal (modulated on carrier freq.)

Digital signal: Represent data with sequence of voltage pulses

- Analog signal ☐CODEC ☐ Digital signal
- Digital signal ☐Digital Transmitter ☐ Digital signal

Data Encoding

- Both analog and digital information can be encoded as either analog or digital signals.

- The particular encoding that **is chosen depends on the specific requirements to be met and the media** and communications facilities available.

Encoding Techniques

- How to represent data in terms of signals?
- Techniques take the **type of data** then to represent via a digital or analog signal
- Four techniques for Data Encoding and Transmission
  - Digital data over digital signal (complex)
  - Digital data over analog signal (switching equipments)
  - Analog data over digital signal (Fiber optics, Unguided media)
  - Analog data over analog signal (voice grade)

Data Encoding and Transmission

- Digital data transmission over digital signal
  - NRZ encoding, Multilevel binary encodings, Biphase encodings, Scrambling techniques
- Digital data transmission over analog signal
  - Public telephone system, Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK), Performance of digital to analog modulation schemes, Quadrature Amplitude Modulation (QAM)
- Analog data transmission over digital signal
  - Digitization, Pulse Code Modulation (PCM), Non-linear encoding, Delta modulation
- Analog data transmission over analog signal
  - Asynchronous transmission, Synchronous transmission, Ethernet link layer frame example

Encoding Techniques

- **Digital data, digital signal**
  - The simplest form is to assign one voltage level to binary one and another to binary zero.
  - More complex encoding schemes are used to improve performance, by altering the spectrum of the signal and providing synchronization capability.
- **Digital data, analog signal**
  - A modem converts digital data to an analog signal so that it can be transmitted over an analog line.
  - The basic techniques are amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK).
    - All involve altering one or more characteristics of a carrier signal to represent binary data

Encoding Techniques

- **Analog data, digital signal**
  - Analog data, such as voice and video, are often digitized to be able to use digital transmission facilities.
  - technique is pulse code modulation (PCM), which involves sampling the analog data periodically and quantizing the samples.
- **Analog data, analog signal**
  - Analog data are **modulated** by a carrier frequency to produce an analog signal in a different frequency band
  - The basic techniques are amplitude modulation (AM), frequency modulation (FM), and phase modulation (PM).

Digital Signal, Analog Signal

- Digital signal
  - Discrete, **discontinuous voltage pulses**
  - Each pulse is a signal element
  - Binary data encoded into signal elements
- Analog signal
  - basis is a **continuous constant-frequency signal/pulse** known as the **carrier signal**.
  - The frequency of the carrier signal is chosen to be **compatible with the transmission medium** being used.
  - Data may be transmitted using a carrier signal by modulation.

Encoding : Refers to Digital Signal as carrier
Modulation: Refers to Analog Signal as carrier

Encoding and Modulation

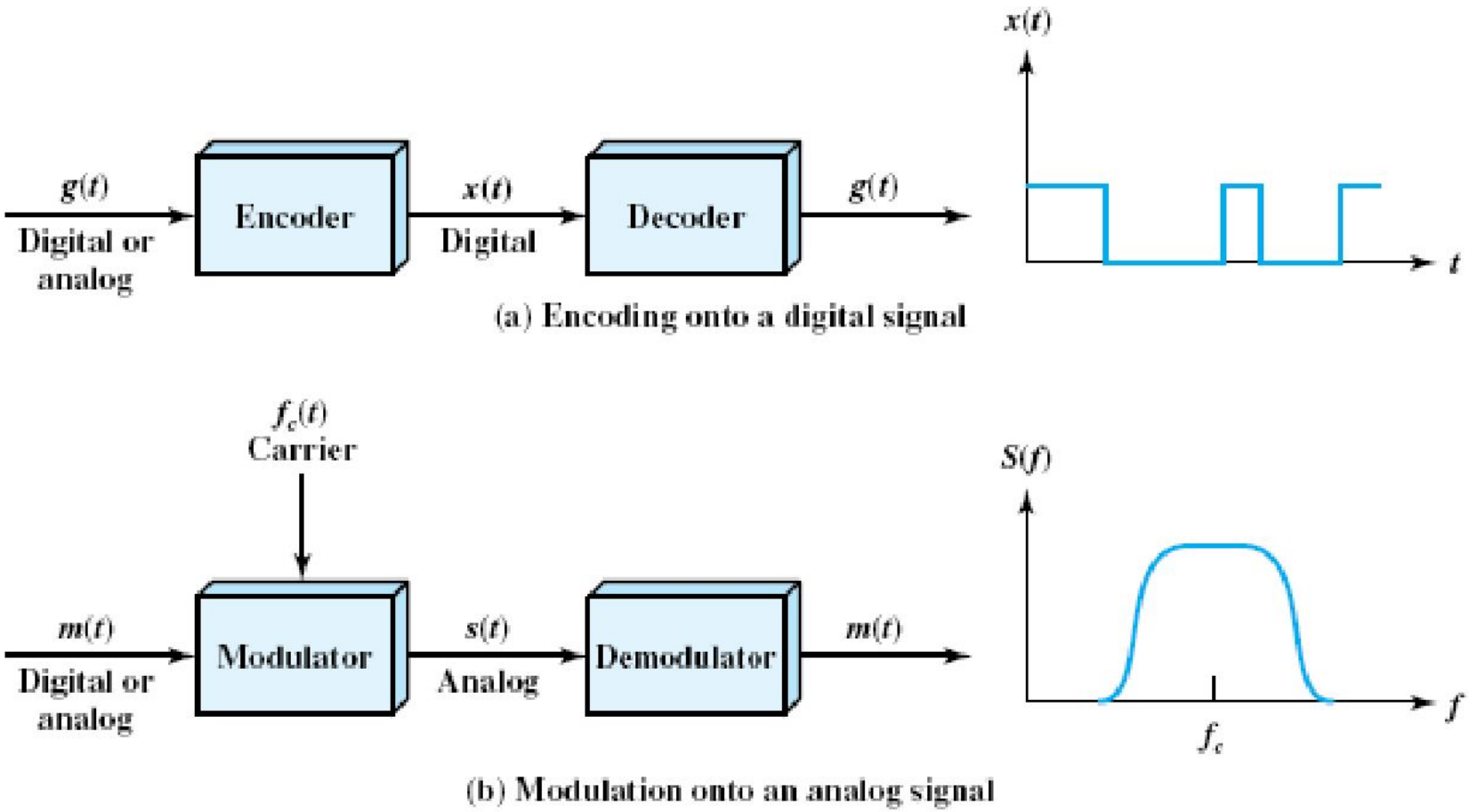


Figure 5.1 Encoding and Modulation Techniques

m(t) is modulating signal or **baseband signal (band-limited signal)**

Modulation

- Modulation is the process of encoding source data onto a carrier signal (Analog carrier) with frequency f_c
- All modulation techniques involve operation on one or more of the three fundamental frequency domain parameters:
  - amplitude, frequency, and phase.

Signal Element

Digital:

- A digital signal is a sequence of discrete, discontinuous **voltage pulses of constant amplitude**. Each pulse is a signal element.
- Binary data are transmitted by encoding each data bit into signal elements.
- In the simplest case, there is a one-to-one correspondence between bits and signal elements.

Analog:

- A **pulse of constant frequency, phase and amplitude**

Key Data Transmission Terms

Term	Units	Definition
Data element	Bits	A single binary one or zero
Data rate	Bits per second (bps)	The rate at which data elements are transmitted
Signal element	Digital: a voltage pulse of constant amplitude	That part of a signal that occupies the shortest interval of a signaling code
	Analog: a pulse of constant frequency, phase, and amplitude	
Signaling rate or modulation rate	Signal elements per second (baud)	The rate at which signal elements are transmitted

Terms (1)

- Unipolar
  - All signal elements have same sign.
    - If the signal elements all have the same algebraic sign, that is, all positive or negative, then the signal is **unipolar.**
- Polar
  - One logic state represented by positive voltage and the other by negative voltage
- Data rate/Bit rate
  - Rate of data transmission in bits per second
- Duration or length of a bit
  - Time taken for transmitter to emit the bit

Terms (2)

- Modulation rate
  - Rate at which the signal level changes
  - Measured in **baud** = signal elements per second
    - i.e. the number of samples/sec made
- Mark and Space
  - Binary 1 and Binary 0 respectively

Note

Bit rate, N, is the number of bits per second (bps). Baud rate is the number of signal elements per second (bauds).

In the analog transmission of digital data, the signal or baud rate is less than or equal to the bit rate.

$S = N \times 1/r$ bauds

Where, r is the number of data bits per signal element.

Example

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

Solution

In this case, r = 4, S = 1000, and N is unknown. We can find the value of N from

$$S = N \times \frac{1}{r} \qquad \text{or} \qquad N = S \times r = 1000 \times 4 = 4000 \text{ bps}$$

Example

An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

Solution

In this example, S = 1000, N = 8000, and r and L are unknown. We find first the value of r and then the value of L.

$$S = N \times \frac{1}{r} \rightarrow r = \frac{N}{S} = \frac{8000}{1000} = 8 \text{ bits/baud}$$

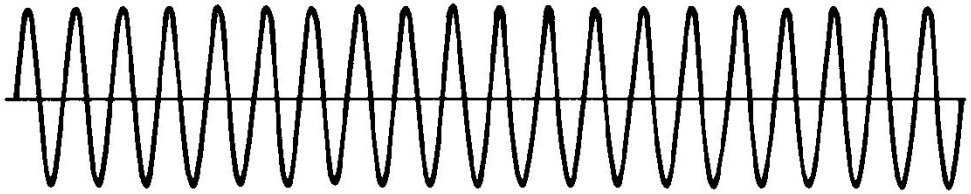
$$r = \log_2 L \rightarrow L = 2^r = 2^8 = 256$$

Analog Data, Analog Signal

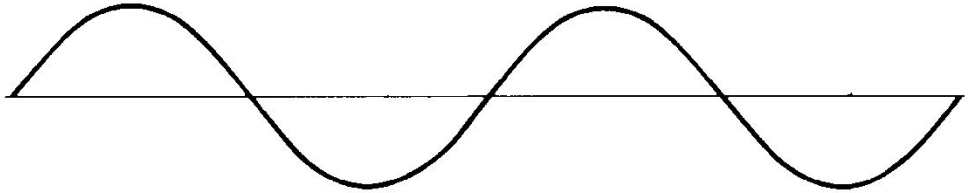
Analog Data, Analog Signals

- Baseband Transmission
  - If signals are **transmitted at their original spectrum** is referred to as baseband transmission.
  - Example :
    - voice signals are transmitted over telephone lines at their original spectrum
- When the data are already analog, is less clear for transmission.
- Why modulate analog signals?
  - **Higher frequency can give more efficient transmission.**
  - For unguided transmission, it is virtually impossible to transmit baseband signals; the required antennas would be many kilometers in diameter.
  - **Permits FDM (frequency division multiplexing)** (chapter 8), which provides efficiency.
- Types of modulation
  - Amplitude
  - Frequency
  - Phase

Analog Modulation

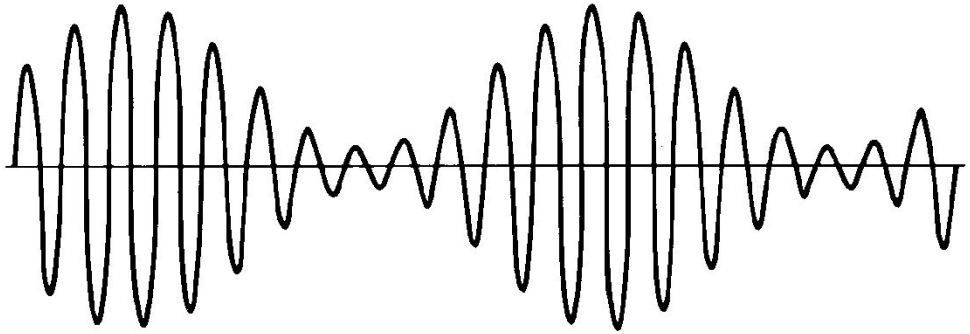


Carrier



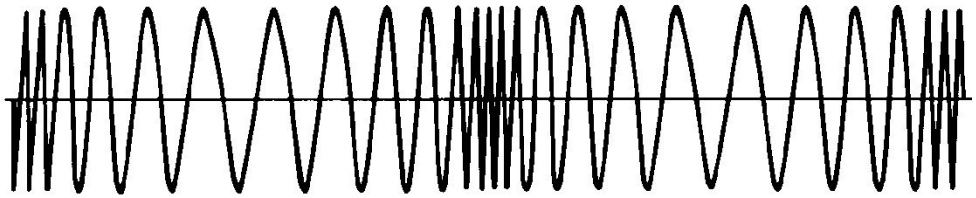
Modulating sine-wave signal

Amplitude of the carrier is modulated (Double side band carrier)



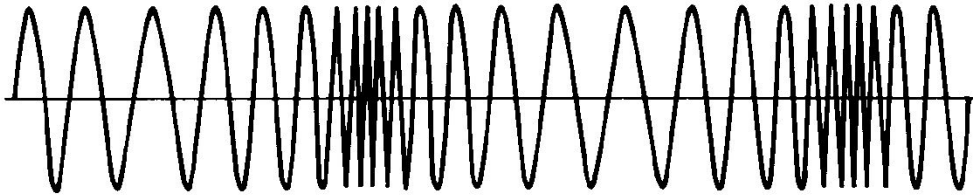
Amplitude-modulated (DSBTC) wave

Phase of the carrier is modulated



Phase-modulated wave

Frequency of the carrier is modulated



Frequency-modulated wave

Amplitude Modulation (AM)

- Amplitude modulation (AM) is the simplest form of modulation. Mathematically, the process can be expressed as

$$\text{AM} \qquad s(t) = [1 + n_a x(t)]\cos 2\pi f_c t$$

'1' is DC component that prevents loss of information

Modulation Index : the ratio of the amplitude of the input signal to the carrier

Input signal

Carrier signal

- Therefore, the input signal will be, $m(t) = n_a x(t)$
- This scheme is also known as DSB-TC (Double SideBand Transmitted Carrier).
- AM involves the multiplication of the input signal by carrier

Amplitude Modulation (AM)

EXAMPLE 5.4 Derive an expression for $s(t)$ if $x(t)$ is the amplitude-modulating signal $\cos 2\pi f_m t$. We have

$$s(t) = [1 + n_a \cos 2\pi f_m t]\cos 2\pi f_c t$$

By trigonometric identity, this may be expanded to

$$s(t) = \cos 2\pi f_c t + \frac{n_a}{2}\cos 2\pi(f_c - f_m)t + \frac{n_a}{2}\cos 2\pi(f_c + f_m)t$$

The resulting signal has a component at the original carrier frequency plus a pair of components each spaced f_m hertz from the carrier.

The envelope of the resulting signal : $[1+ n_a x(t)]$

For $n_a < 1$: envelope is exact reproduction of original

For $n_a > 1$: envelope will cross the time axis and information is lost

Amplitude Modulation (AM)

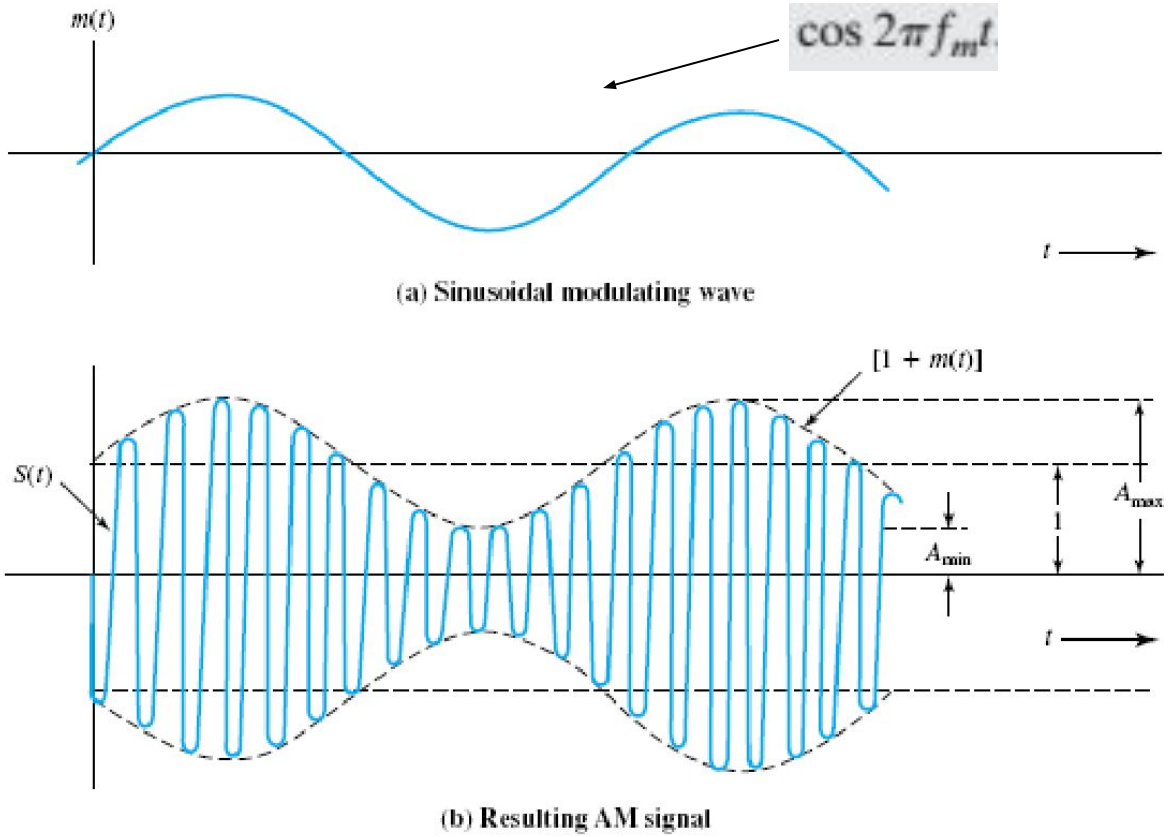


Figure 5.22 Amplitude Modulation

$$s(t) = [1 + n_a \cos 2\pi f_m t]\cos 2\pi f_c t$$

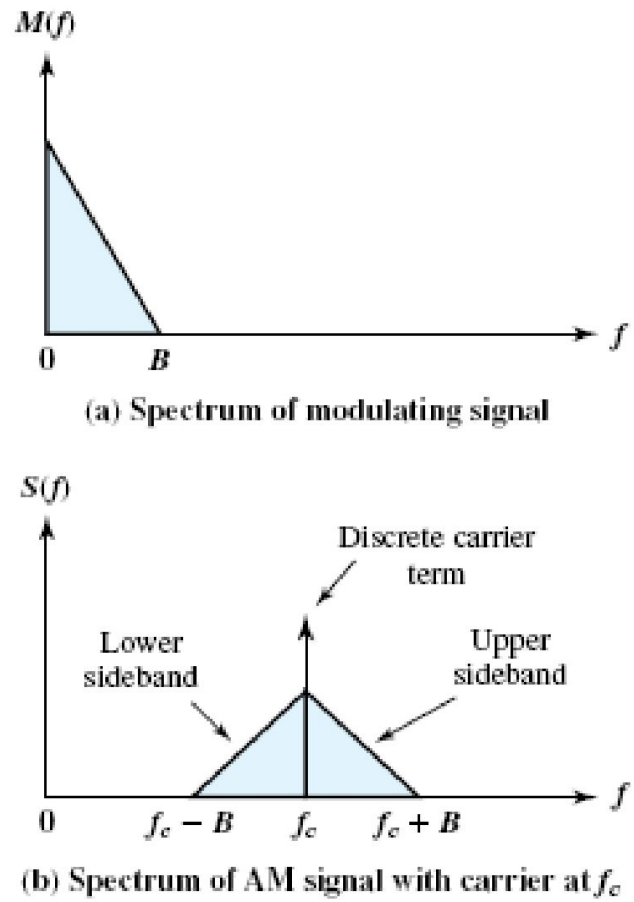
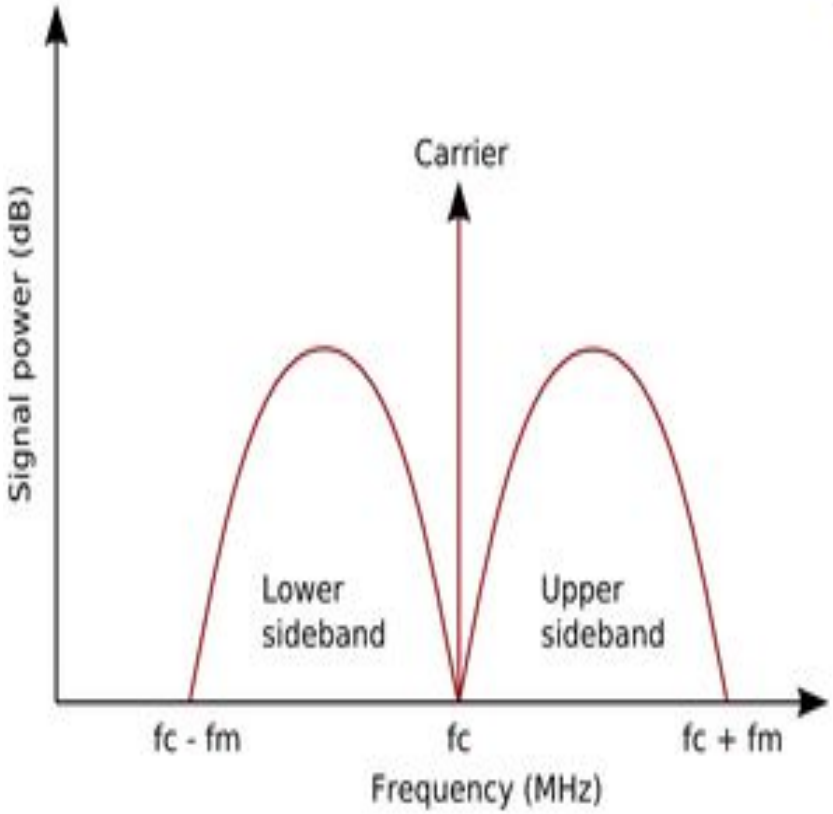


Figure 5.23 Spectrum of an AM Signal

Amplitude Modulation (AM)

- Amplitude modulationAmplitude modulation of a carrier wave normally results in two mirror-image sidebands.
- The signal components above the carrier frequency constitute the **upper sideband** (**USB**),
- And below the carrier frequency constitute the **lower sideband** (**LSB**).
- In conventional AM transmission, the carrier and both sidebands are present, sometimes called **double sideband** amplitude modulation (**DSB-AM**).



Variants of AM

- ***s*(*t*) contains unnecessary components**, because each of the sidebands contains the complete spectrum of ***m*(*t*)**.
- Variants of AM (or DSB-AM):
  - **Double sideband Transmitted Carrier (DSB-TC) :** Both the sidebands are transmitted.
  - **Single sideband (SSB) :** takes advantage of this fact by sending only one of the sidebands, eliminating the other sideband and the carrier. The principal advantages of this approach are as follows:
    - Only half of the bandwidth is required, (i.e. B the bandwidth of the original signal only). For DSB-TC, $B_T=2B$
    - Less power is required because no power is used to transmit the carrier or the other sideband.

Variants of AM

- **Double sideband suppressed carrier (DSB-SC) :** It filters out the carrier frequency and sends both sidebands. This saves some power but uses as much bandwidth as DSB-TC.

  The disadvantage of suppressing the carrier is that the carrier can be used for synchronization purposes.

  Example : Stereophonic FM Broadcasting on 38KHz sub-carrier

- **Vestigial sideband (VSB) :** The disadvantage of DSB-SC is addressed with a compromised approach, that is VSB, which uses one sideband and reduced-power carrier.

  If part of one sideband and all of the other remain, it is called vestigial sideband,

  Example: used mostly with TV broadcasting

Angle Modulation (FM and PM)

- Basically Frequency modulation (FM) and phase modulation (PM) are special cases of angle modulation. The modulated signal is expressed as :

Angle Modulation $$s(t) = A_c \cos[2\pi f_c t + \phi(t)]$$

- For Phase modulation, the phase is proportional to the modulating signal:

PM $$\phi(t) = n_p m(t)$$

Phase modulation index

- For frequency modulation, the derivative of the phase is proportional to the modulating signal:

FM $$\phi'(t) = n_f m(t)$$

Derivatives of $\phi(t)$

Frequency modulation index

Angle Modulation (FM and PM)

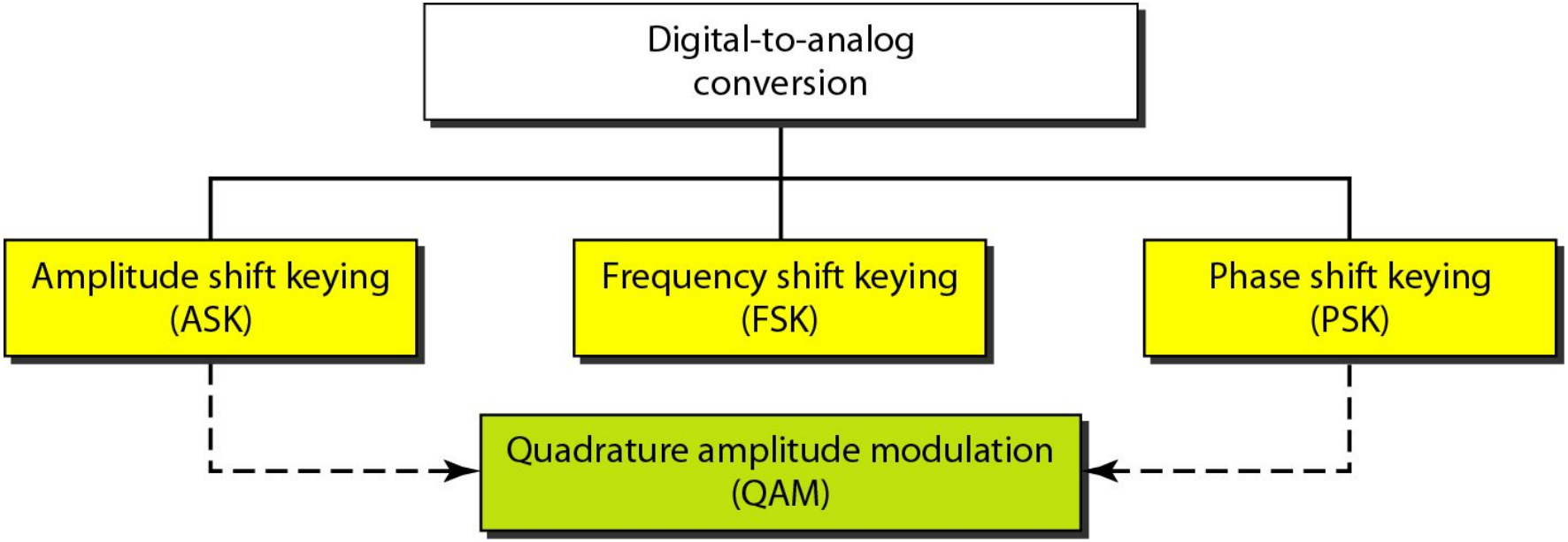
- Phase Modulation Index (PHI):
  - also known as the modulation depth,
  - quantifies how much the modulated variable of the carrier signal varies around its unmodulated level
  - Dictates to which the carrier's phase is shifted by the modulating signal
  - **Higher PMI:** means the phase variations are more sensitive to the baseband signal, resulting in a larger phase shift
  - **Lower PMI:** means the phase variations are less sensitive, leading to smaller phase shifts
  - PM, with its PMI, is used in : satellite communication, GSM, WiFi, and sound synthesizing applications etc.
- Frequency Modulation Index (FMI):
  - the ratio of frequency deviation to the modulating frequency, determines the bandwidth and quality of the transmitted signal

Digital Data, Analog Signal

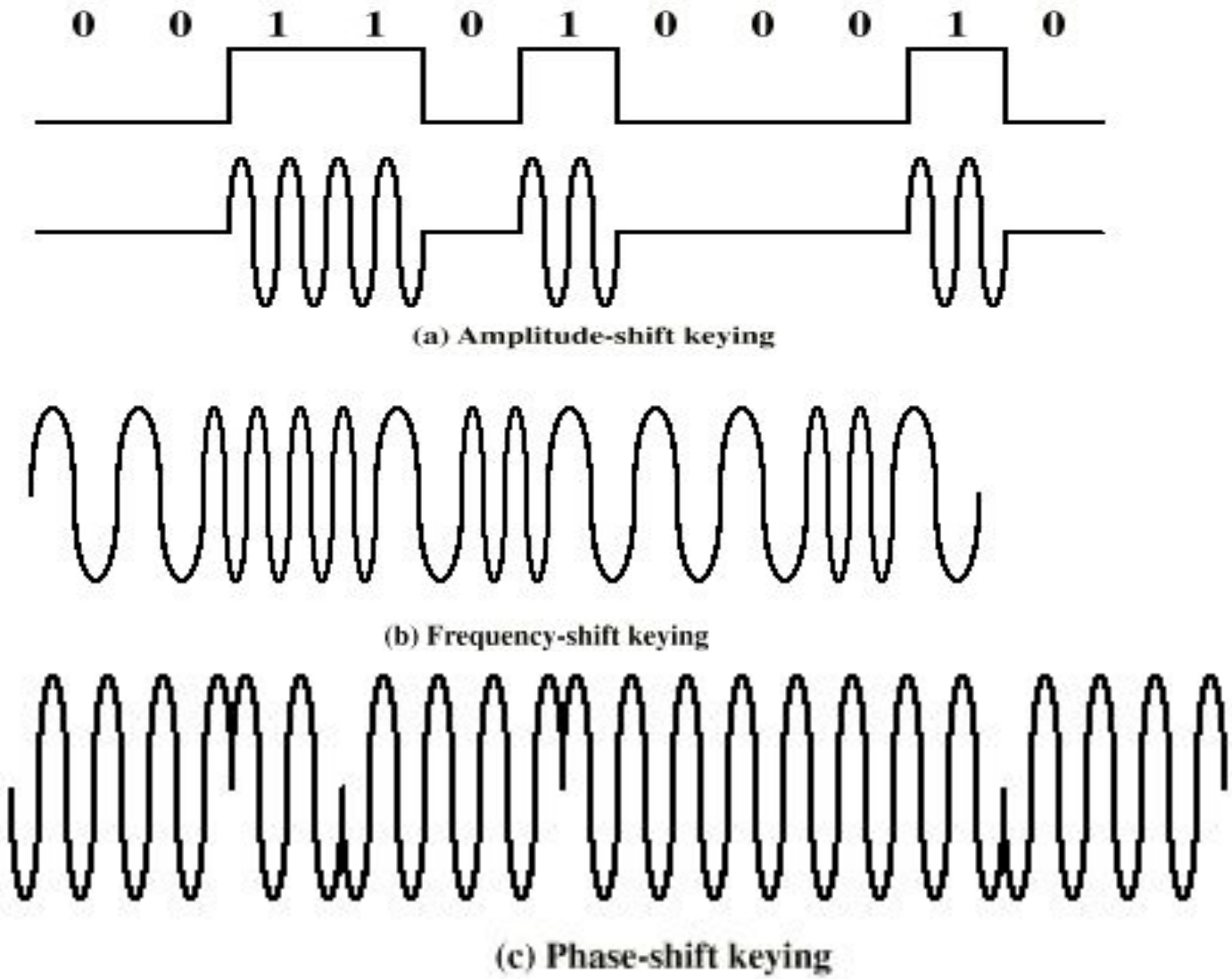
Digital Data, Analog Signal

- Public telephone system
  - 300Hz to 3400Hz
  - Use modem (modulator-demodulator)
- Modulation
  - Amplitude shift keying (ASK)
  - Frequency shift keying (FSK)
  - Phase shift keying (PSK)

Types of digital-to-analog conversion



Modulation Techniques



Modulation

- The modulation technique (e.g., QPSK) determines the number of bits/symbol.
- The bit rate is the amount of information sent over the channel and is equal to the number of symbols/sec times the number of bits/symbol.
- A combination of modulation techniques are used to transmit multiple bits per baud to achieve higher speed (data rate).
  - Often multiple amplitudes and multiple phase shifts are combined to transmit several bits/symbol.

Amplitude Shift Keying

- Two binary Values are represented by different amplitudes of carrier frequency.
- Usually, one amplitude is zero
  - i.e. presence and absence of carrier is used
- Susceptible to sudden gain changes
- Inefficient modulation technique
- Up to 1200bps on voice grade lines
- Used over optical fiber

Amplitude Shift Keying

- The resulting transmitted signal for one bit time is :

$$\textbf{ASK} \qquad s(t) = \begin{cases} A\cos(2\pi f_c t) & \text{binary 1} \\ 0 & \text{binary 0} \end{cases}$$

carrier signal

Frequency Shift Keying

- Common form of FSK is binary FSK (BFSK), in which the two binary values are represented by two different frequencies near the carrier frequency
- Values represented by different frequencies (near carrier)
- Less susceptible to error than ASK
- Up to 1200bps on voice grade lines
- Used for High frequency radio transmission
- Even higher frequency on LANs using co-ax

Frequency Shift Keying

- The resulting transmitted signal for one bit time is :

$$\textbf{BFSK} \qquad s(t) = \begin{cases} A\cos(2\pi f_1 t) & \text{binary } 1 \\ A\cos(2\pi f_2 t) & \text{binary } 0 \end{cases}$$

- f1 and f2 are typically offset from the carrier frequency fc by equal but opposite amounts.

FSK on Voice Grade Line

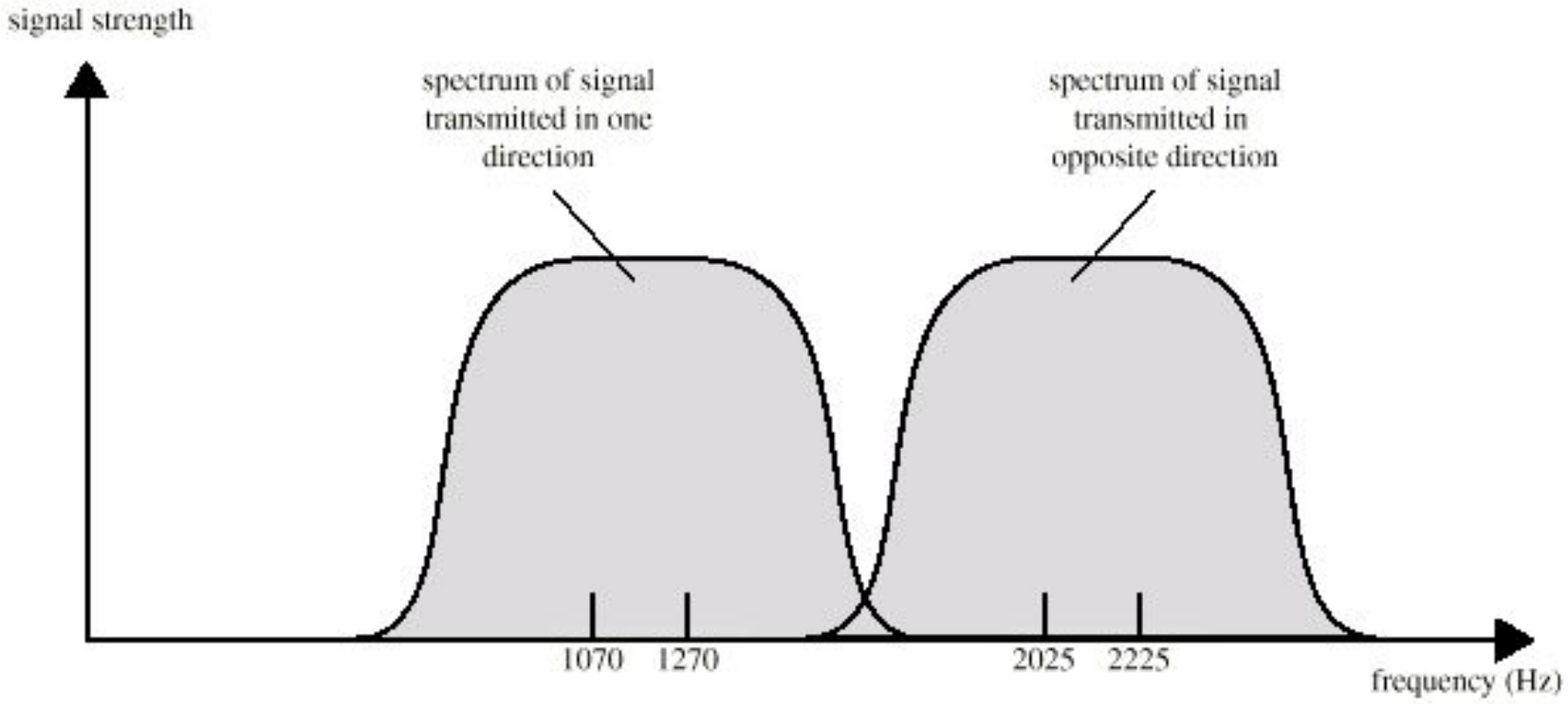


Figure 5.8 Full-Duplex FSK Transmission on a Voice-Grade Line

MFSK

- A signal that is more bandwidth efficient, but also more susceptible to error, is **multiple FSK (MFSK)**, in which more than two frequencies are used. In this case each signaling element represents more than one bit. The transmitted MFSK signal for one signal element time can be defined as follows:

$$\textbf{MFSK} \qquad s_i(t) = A\cos 2\pi f_i t, \qquad 1 \le i \le M$$

where

$$f_i = f_c + (2i - 1 - M)f_d$$
$$f_c = \text{the carrier frequency}$$
$$f_d = \text{the difference frequency}$$
$$M = \text{number of different signal elements} = 2^L$$
$$L = \text{number of bits per signal element}$$

Phase Shift Keying

- Phase of carrier signal is shifted to represent data
- Two-level PSK (BPSK)
  - uses two phases (in terms of angle) to represent the two binary digits and is known as binary phase shift keying.

$$\textbf{BPSK} \qquad s(t) = \begin{cases} A\cos(2\pi f_c t) \\ A\cos(2\pi f_c t + \pi) \end{cases} = \begin{cases} A\cos(2\pi f_c t) & \text{binary } 1 \\ -A\cos(2\pi f_c t) & \text{binary } 0 \end{cases}$$

- Phase is 180^0 always in BPSK.
- An alternative form of two-level PSK is Differential PSK (DPSK).
- DPSK
  - Phase shifted relative to previous signal element rather than some reference signal
  - The signal phase follows the high or low state of the previous element

Quadrature PSK

- Four-Level PSK
  - to make efficient use of BW by allowing more than one bit by each signaling element
- More efficient use by each signal element representing more than one bit
  - e.g. shifts of π/2 (90°)
  - Each element represents two bits
  - Can use 8 phase angles and have more than one amplitude
  - 9600bps modem uses 12 angles , four of which have two amplitudes
- Instead of a phase shift of 180°, as allowed in BPSK, a common encoding technique, known as quadrature phase shift keying (QPSK), uses phase shifts separated by multiples of 90^0

QPSK

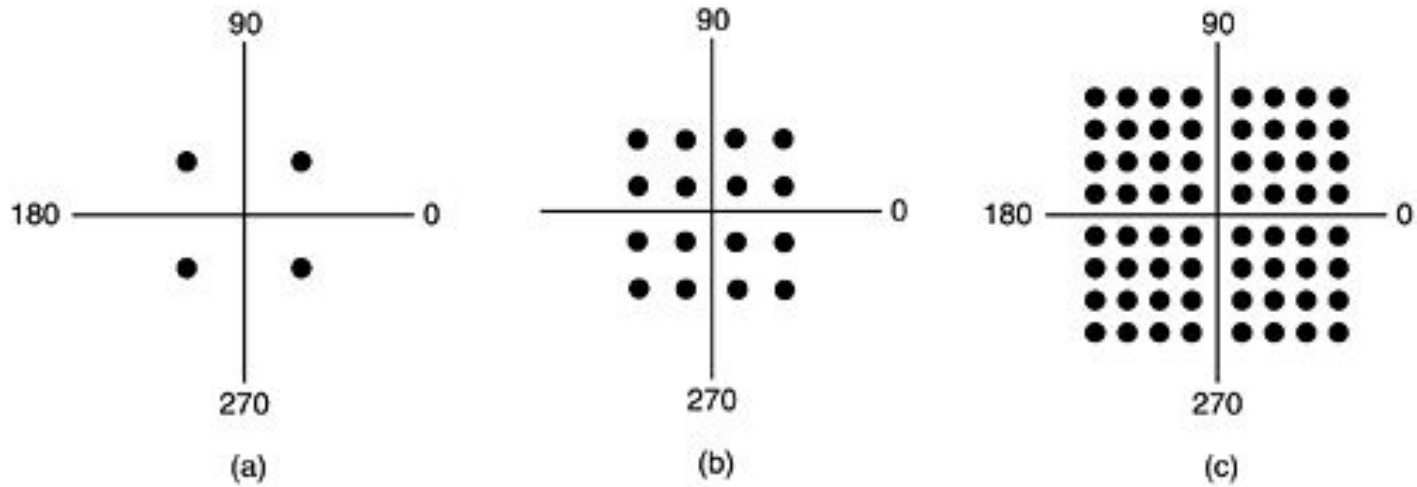
$$\textbf{QPSK} \qquad s(t) = \begin{cases} A\cos\left(2\pi f_c t + \dfrac{\pi}{4}\right) & 11 \\ A\cos\left(2\pi f_c t + \dfrac{3\pi}{4}\right) & 01 \\ A\cos\left(2\pi f_c t - \dfrac{3\pi}{4}\right) & 00 \\ A\cos\left(2\pi f_c t - \dfrac{\pi}{4}\right) & 10 \end{cases}$$

Thus each signal element represents two bits rather than one.

QAM (Quadrature Amplitude Modulation)

- QAM is both an analog and a digital modulation scheme.
  - Constellating the Amplitude and Phase
- It conveys two analog message signals, or two digital bit streams, by changing (*modulating*) the amplitudes) the amplitudes of two carrier waves, using the ASK digital modulation scheme or AM analog modulation scheme.
- The two carrier waves (two copies each modulated with ASK)), usually sinusoids, are out of phasewaves (two copies each modulated with ASK)), usually sinusoids, are out of phase with each other by 90° and are thus called quadrature carriers or quadrature components — hence the name of the scheme is QAM.
- The modulated waves are summed, and the resulting waveform is a combination of both PSK and ASK, or (in the analog case) of phase modulation (PM) and amplitude modulation (AM).

QPSK, QAM-16, QAM-64



- The phase of a dot is indicated by the angle a line from it to the origin makes with the positive x-axis.
- Fig(a) has four valid combinations and can be used to transmit 2 bits per symbol.

QPSK, QAM-16, QAM-64

- In Fig.(b) QAM scheme is used, in which four amplitudes and four phases are used , for a total of 16 different combinations.
  - This modulation scheme can be used to transmit 4 bits per symbol.
  - It is called QAM-16 (Quadrature Amplitude Modulation). Sometimes the term 16-QAM is used instead.
  - QAM-16 can be used, for example, to transmit 9600 bps over a 2400-baud line.
- Fig.(c) is another modulation scheme involving amplitude and phase.
  - It allows 64 different combinations, so 6 bits can be transmitted per symbol. It is called QAM-64. Higher-order QAMs also are used.

Constellation

- Diagrams such as those of Fig.s, which show the legal **combinations of amplitude and phase,** are called constellation diagrams.
- Constellation and Constellation Pattern
  - Technique of using combination of amplitude and phase is called constellation
  - The representation of the bits through constellation is called the constellation pattern
- Each high-speed modem standard has its own constellation pattern and can talk only to other modems that use the same one

Performance of Digital to Analog Modulation Schemes

- Bandwidth
  - ASK and PSK bandwidth is directly **related to bit rate**
  - FSK bandwidth is **related to data rate** for lower frequencies, but to offset of modulated frequency from carrier at high frequencies
  - (See Stallings for math)
- In the presence of noise, bit error rate of PSK and QPSK are about 3dB superior to ASK and FSK

Analog Data, Digital Signal

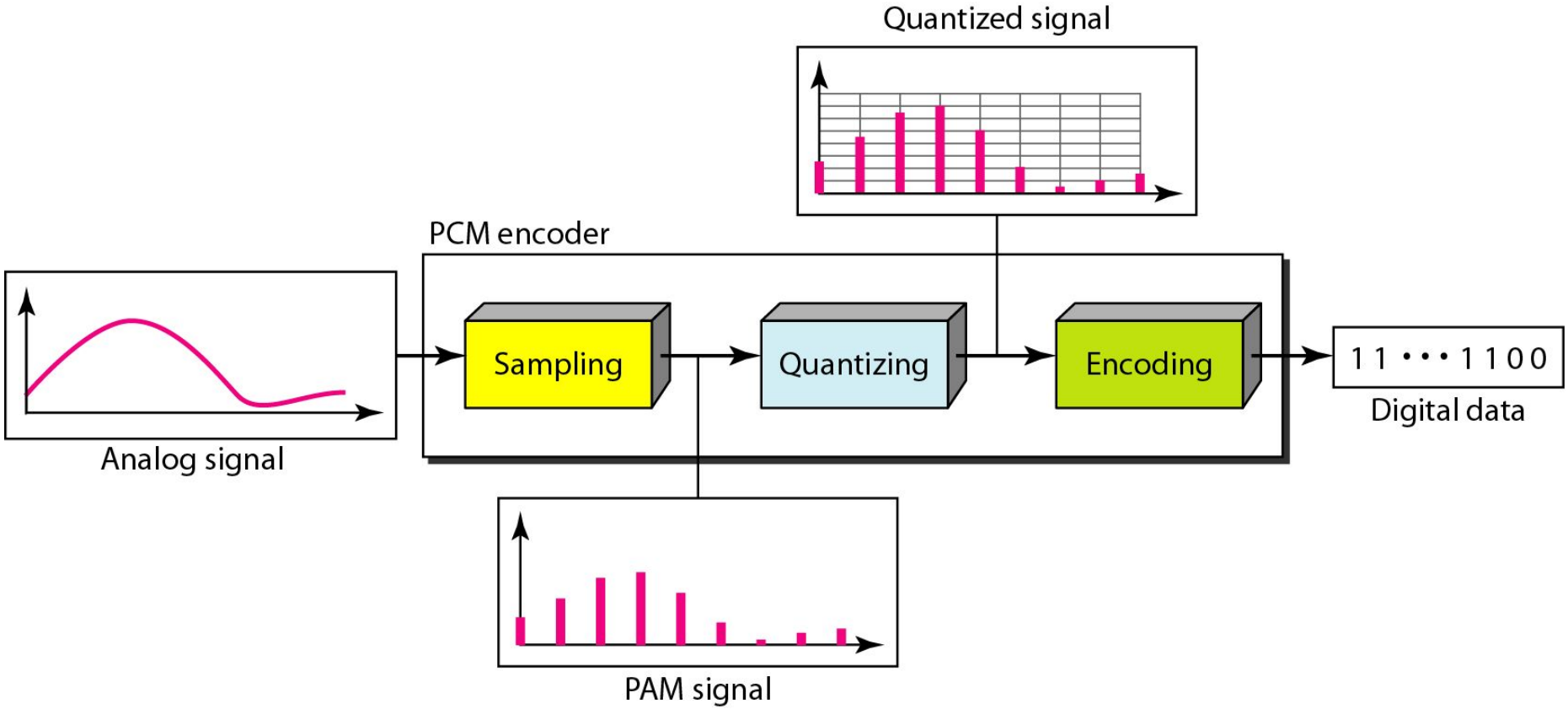
Analog Data, Digital Signal

- This is a process of converting analog data into digital data; this process is known as **digitization**.
- Digitization
  - Conversion of analog data into digital data
  - Digital data can then be transmitted using NRZ-L
  - Digital data can then be transmitted using code other than NRZ-L
  - Digital data can then be converted to analog signal
  - Analog to digital conversion is done using a **codec (coder-decoder)**
- Two techniques
  - Pulse code modulation (PCM) : based on sampling theory
  - Delta modulation (DM)

PCM

- PCM consists of three steps to digitize an analog signal:
  1. Sampling
  2. Quantization
  3. Binary encoding
- Before we sample, we have to filter the signal to limit the maximum frequency of the signal as it affects the sampling rate.
- Filtering should ensure that we do not distort the signal, i.e. remove high frequency components that affect the signal shape.

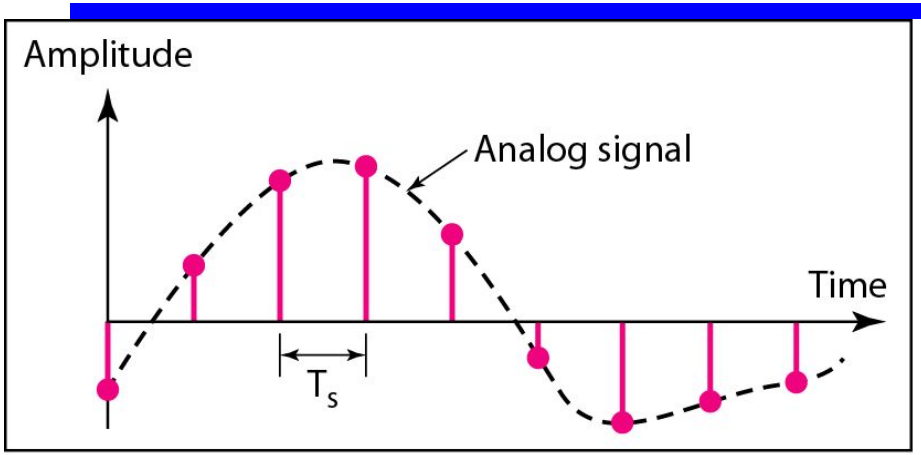
Components of PCM encoder



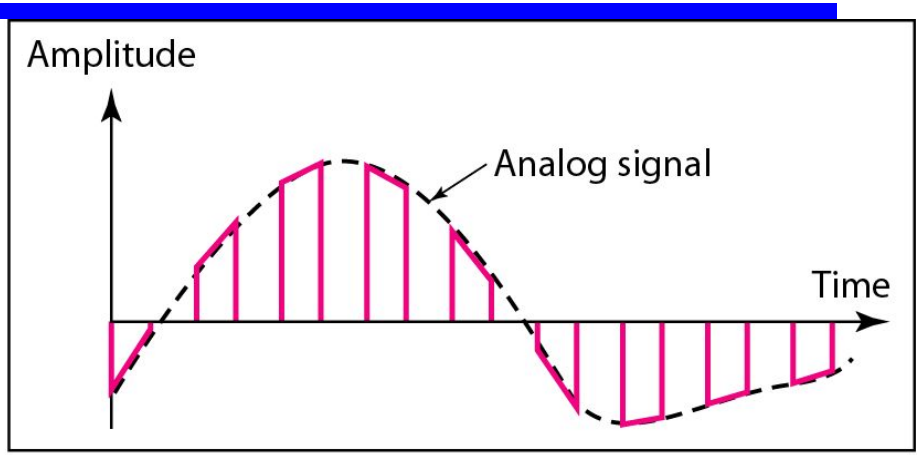
Sampling

- Analog signal is sampled every T_S secs.
- T_s is referred to as the sampling interval.
- $f_s = 1/T_s$ is called the sampling rate or sampling frequency.
- There are 3 sampling methods:
  - Ideal - an impulse at each sampling instant
  - Natural - a pulse of short width with varying amplitude
  - Flattop - sample and hold, like natural but with single amplitude value
- The process is referred to as pulse amplitude modulation PAM and the outcome is a signal with analog (non integer) values

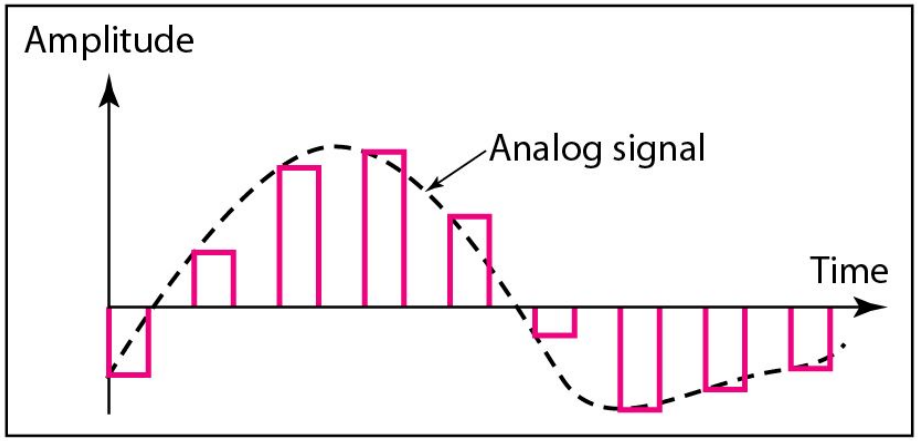
Three different sampling methods for PCM



a. Ideal sampling



b. Natural sampling

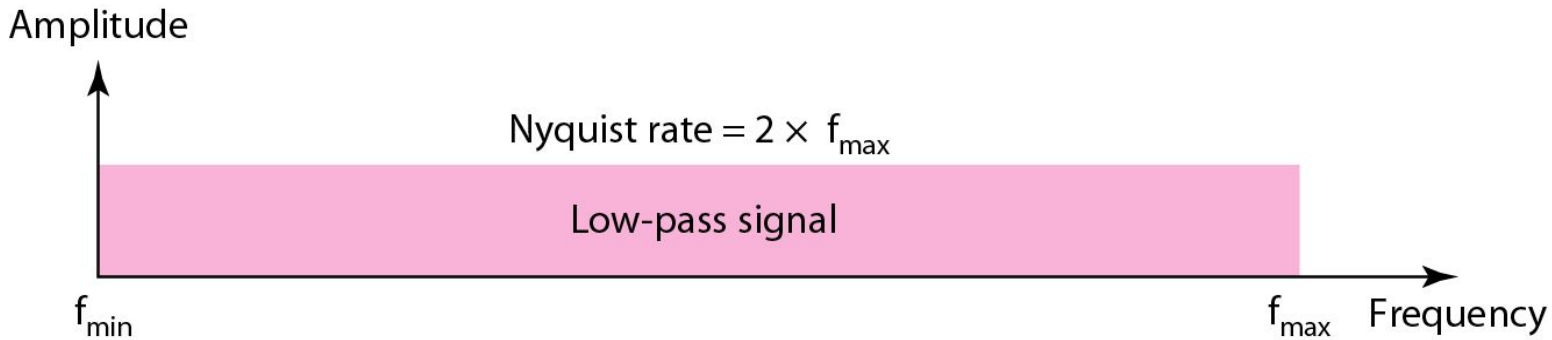


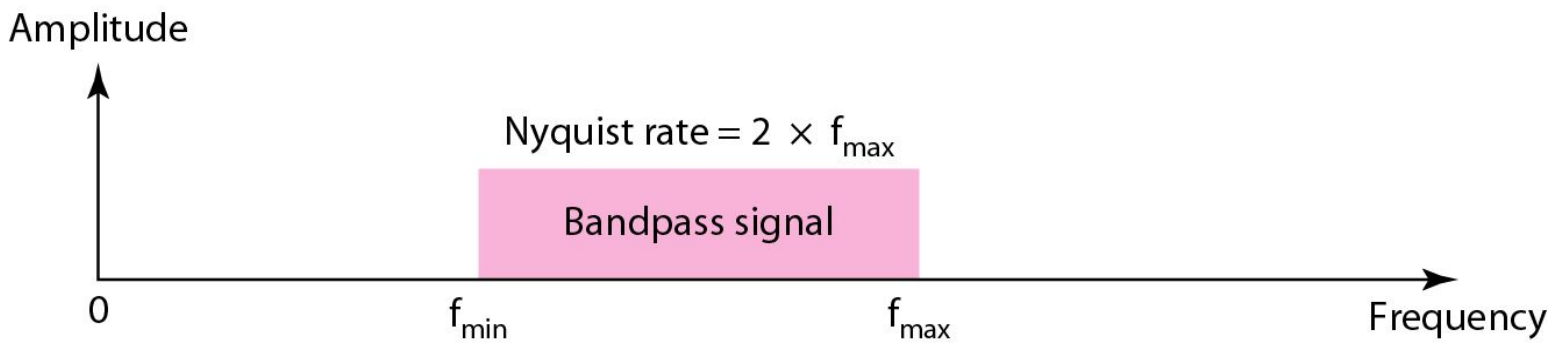
c. Flat-top sampling

Note

According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

Nyquist sampling rate for low-pass and bandpass signals





Pulse Code Modulation(PCM) (1)

- If **a signal f(t) is sampled at regular intervals of time at a rate higher than twice the highest signal frequency**, then the samples contain all the information of the original signal. f(t) is reconstructed using low pass filter.
  - (Proof - Stallings appendix 4A)
- Voice data limited to below 4000Hz, so it require 8000 sample per second
- Analog samples (Pulse Amplitude Modulation, PAM)
- Each sample is assigned digital value
- PAM samples are approximated by Quantization into different levels

Pulse Code Modulation(PCM) (2)

- Say each PAM sample is approximated by being quantized into one of 16 different levels.
  - 4 bit system gives 16 levels
  - Because of quantization, it is impossible to recover the original signal exactly. So, 4 bit system is not enough.
- Quantized issues
  - Quantizing error or noise
  - Approximations mean it is impossible to recover original exactly
- By 8 bit sample gives 256 quantizing levels, the Quality of the recovered voice signal is comparable with analog transmission
- 8000 samples per second of 8 bits each gives 64kbps
- PCM starts with a continuous-time, continuous-amplitude (analog) signal, from which a digital signal is produced
  - Digital signal consists of blocks of n bits, each n bit number is the amplitude of a PCM pulse.

Pulse Code Modulation(PCM) (3)

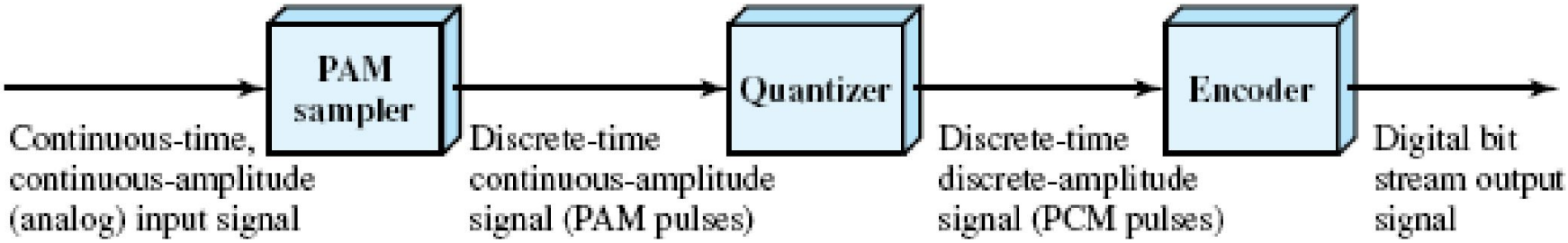
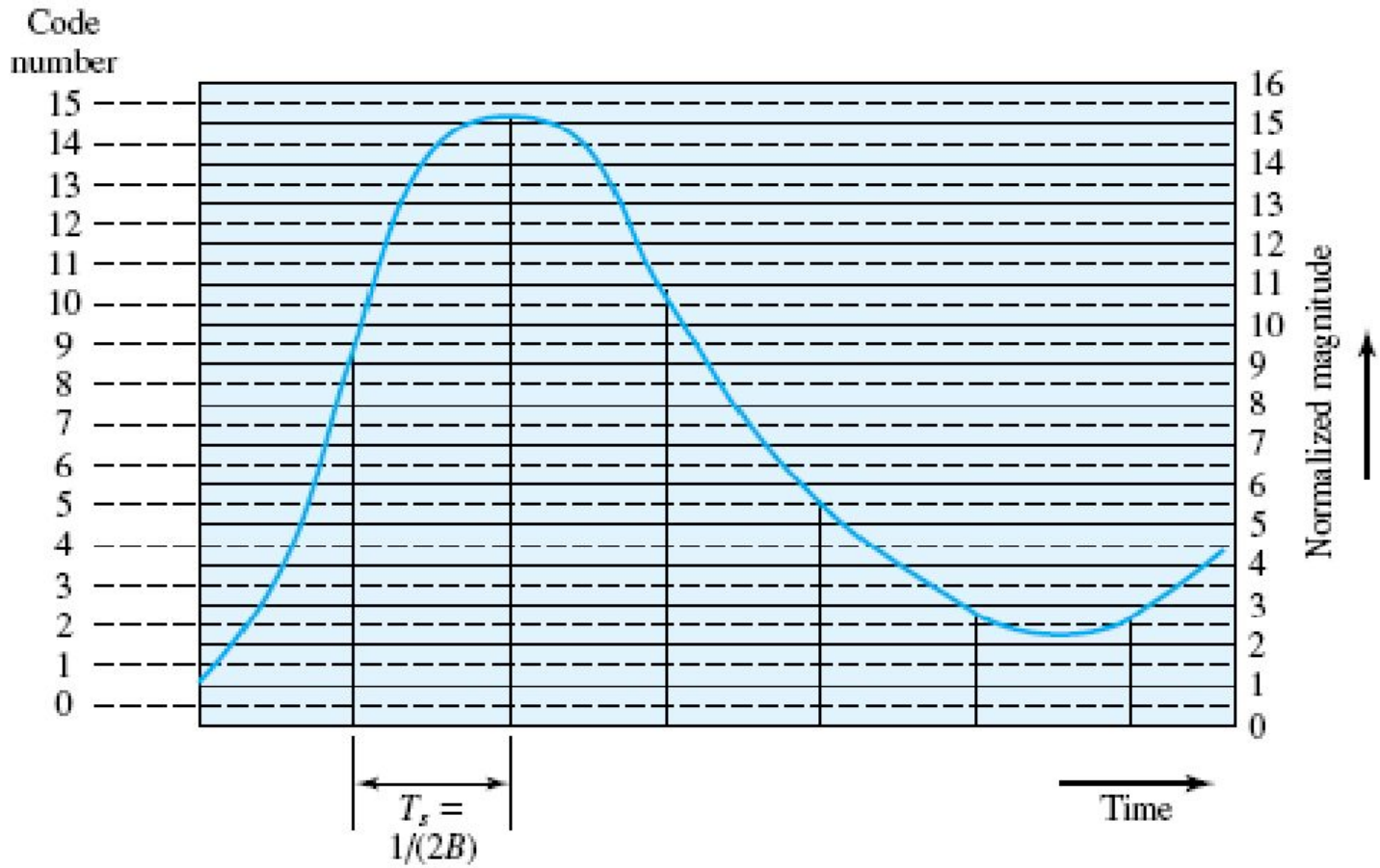


Figure 5.17 PCM Block Diagram

PCM Example



PAM value	1.1	9.2	15.2	10.8	5.6	2.8	2.7
Quantized code number	1	9	15	10	5	2	2
PCM code	0001	1001	1111	1010	0101	0010	0010

Figure 5.16 Pulse Code Modulation Example

Quantization

- Sampling results in a series of pulses of varying amplitude values ranging between two limits: a min and a max.
- The amplitude values are infinite between the two limits.
- We need to map the *infinite* amplitude values onto a finite set of known values.
- This is achieved by dividing the distance between min and max into L zones, each of height Δ.

$$\Delta = (\max - \min)/L$$

Quantization Levels

- The midpoint of each zone is assigned a value from 0 to L-1 (resulting in L values)
- Each sample falling in a zone is then approximated to the value of the midpoint.

Quantization Zones

- Assume we have a voltage signal with amplitutes V_{min}=-20V and V_{max}=+20V.
- We want to use L=8 quantization levels.
- Zone width Δ = (20 - -20)/8 = 5
- The 8 zones are: -20 to -15, -15 to -10, -10 to -5, -5 to 0, 0 to +5, +5 to +10, +10 to +15, +15 to +20
- The midpoints are: -17.5, -12.5, -7.5, -2.5, 2.5, 7.5, 12.5, 17.5

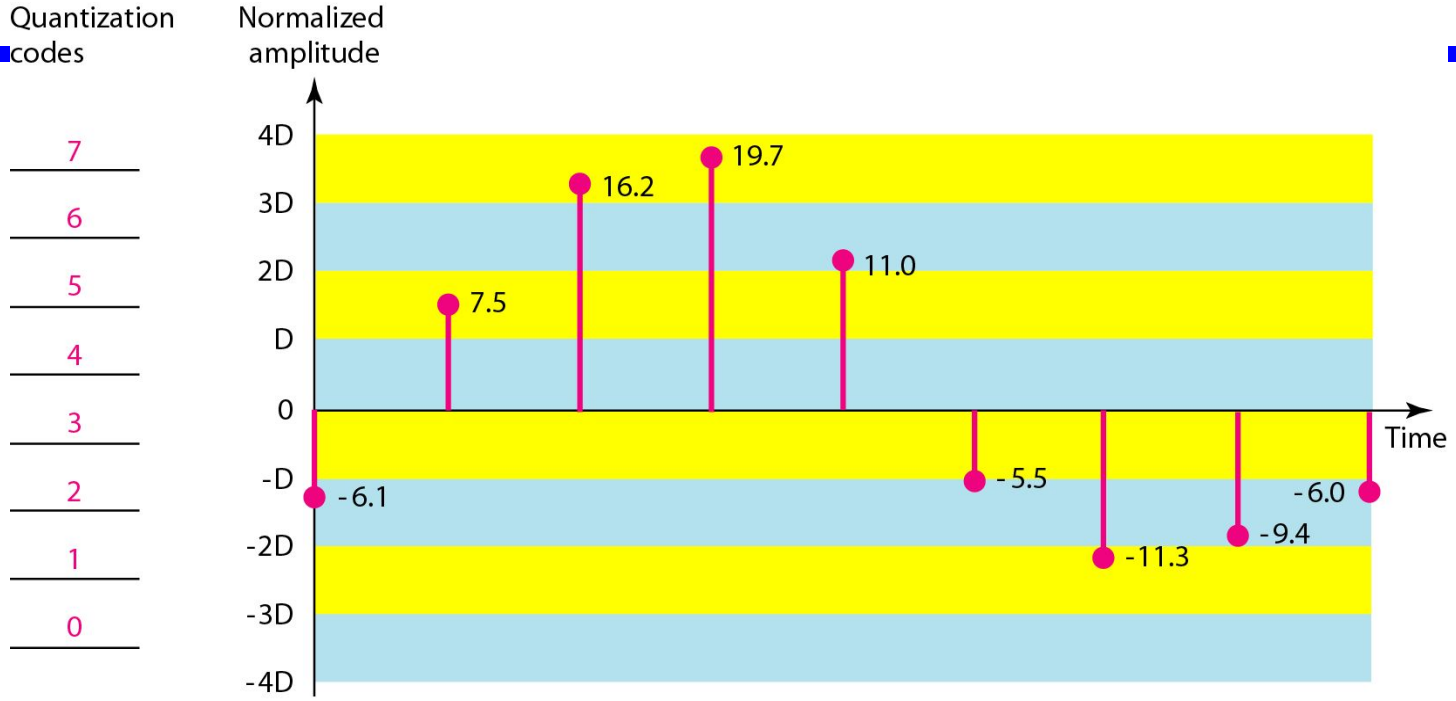
Assigning Codes to Zones

- Each zone is then assigned a binary code.
- The number of bits required to encode the zones, or the number of bits per sample as it is commonly referred to, is obtained as follows:

$$n_b = \log_2 L$$

- Given our example, $n_b = 3$
- The 8 zone (or level) codes are therefore: 000, 001, 010, 011, 100, 101, 110, and 111
- Assigning codes to zones:
  - 000 will refer to zone -20 to -15
  - 001 to zone -15 to -10, etc.

Quantization and encoding of a sampled signal



Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

Quantization Error

- When a signal is quantized, we introduce an error - the coded signal is an approximation of the actual amplitude value.
- The difference between actual and coded value (midpoint) is referred to as the quantization error.
- The more zones, the smaller Δ which results in smaller errors.
- BUT, the more zones the more bits required to encode the samples -> higher bit rate

Quantization Error and SN_QR

- Signals with lower amplitude values will suffer more from quantization error as the error range: $\Delta/2$, is fixed for all signal levels.
- Non linear quantization is used to alleviate this problem. Goal is to keep SN_QR fixed for all sample values.
- Two approaches:
  - The quantization levels follow a logarithmic curve. Smaller Δ's at lower amplitudes and larger Δ's at higher amplitudes.
  - Companding: The sample values are compressed at the sender into logarithmic zones, and then expanded at the receiver. The zones are fixed in height.

  [Companding = compressing-expanding the input analog signal]

Nonlinear Encoding

- Quantizing the PAM pulse the original signal is approximated and can't be recovered exactly □ quantizing error/noise
  - Overcome from this by reefing the PCM scheme called nonlinear encoding
- **i.e. Quantization levels are not equally/evenly spaced (equal spacing means□ absolute error for each sample is same i.e. lower amplitude values are more distorted)**
  - Reduces overall signal distortion
- Can also be done by **companding**
- Companding = compressing-expanding the input analog signal
  - Process of compressing the intensity range of a signal by imparting more gain to weak signals than to strong signals

Bit rate and bandwidth requirements of PCM

- The bit rate of a PCM signal can be calculated from the number of bits per sample x the sampling rate

$$\text{Bit rate} = n_b \times f_s$$

- The bandwidth required to transmit this signal depends on the type of line encoding used.

- A digitized signal will always need more bandwidth than the original analog signal. Price we pay for robustness and other features of digital transmission.

Example

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

Solution

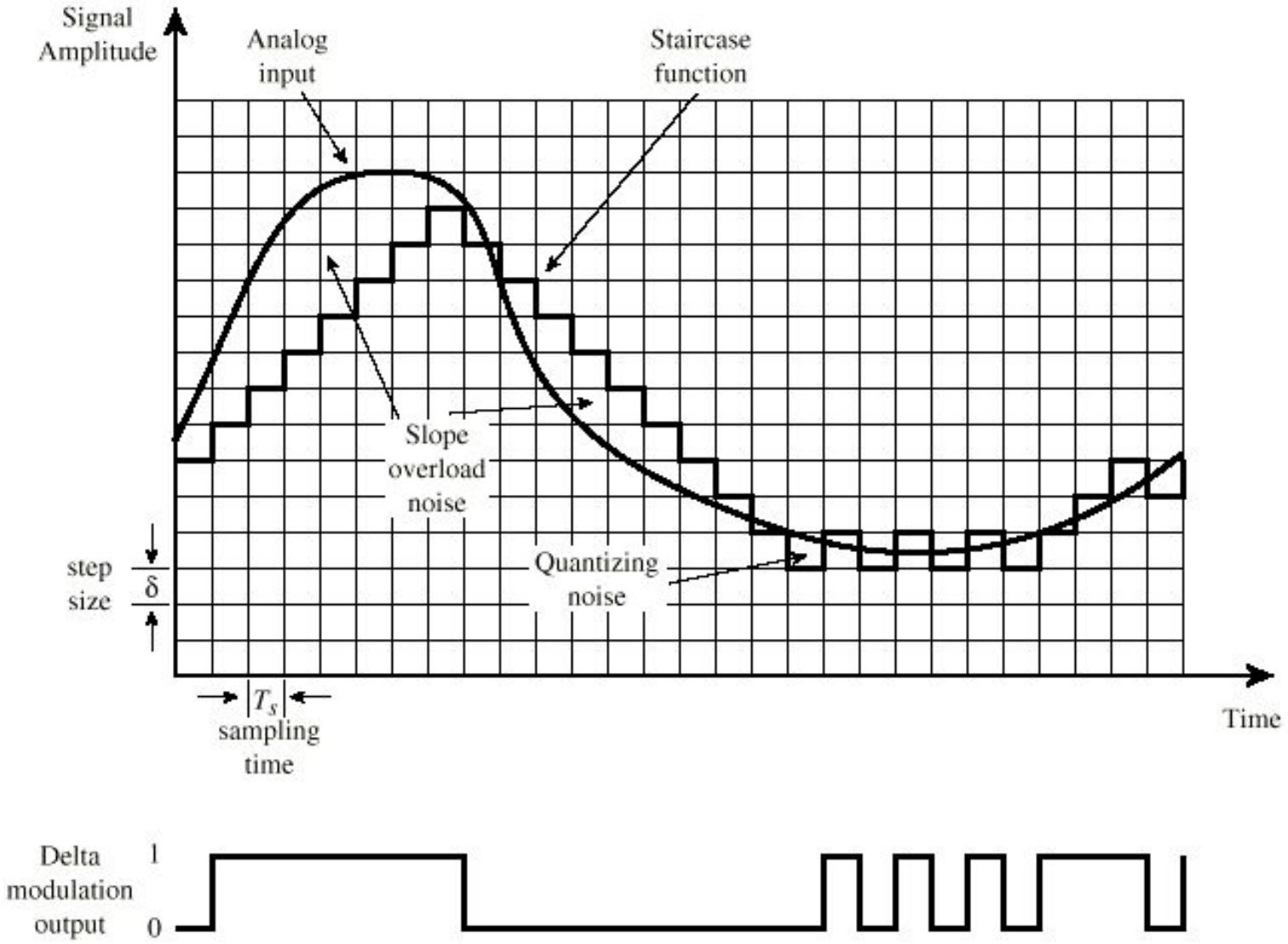
The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows:

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s}$$
$$\text{Bit rate} = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

Delta Modulation (DM)

- Alternative to PCM is DM to improve the performance of PCM.
- Analog input is approximated by a staircase function
- Move up or down one level (δ) at each sample interval
- Binary behavior
  - Function moves up or down at each sample interval

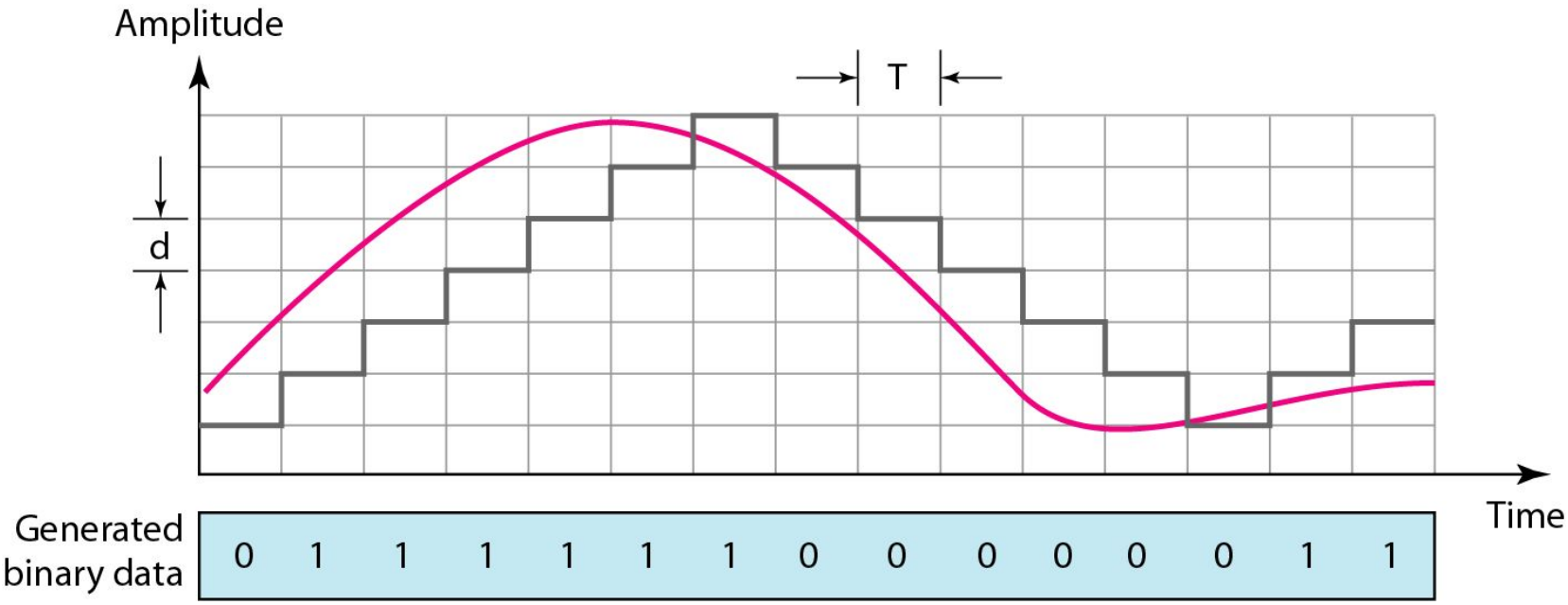
Delta Modulation - example



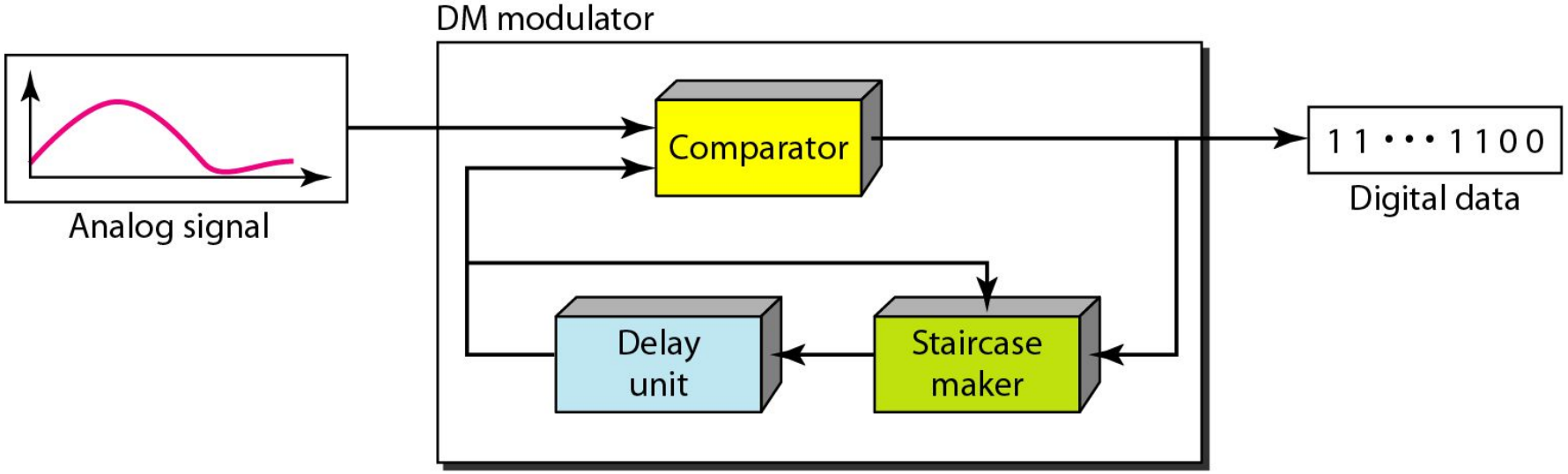
Delta Modulation

- This scheme sends only the difference between pulses,
  - If the pulse at time t_{n+1} is higher in amplitude value than the pulse at time t_n, then a single bit, say a "1", is used to indicate the positive value.
  - If the pulse is lower in value, resulting in a negative value, a "0" is used.
- This scheme works well for small changes in signal values between samples.
- If changes in amplitude are large, this will result in large errors.

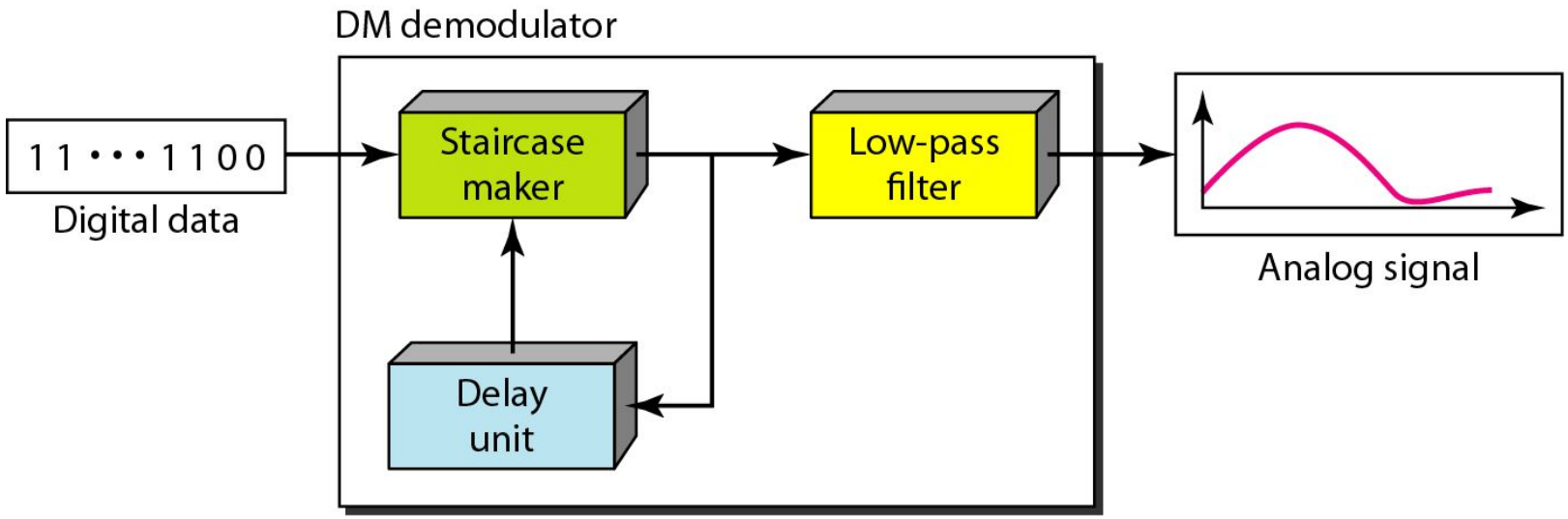
The process of delta modulation



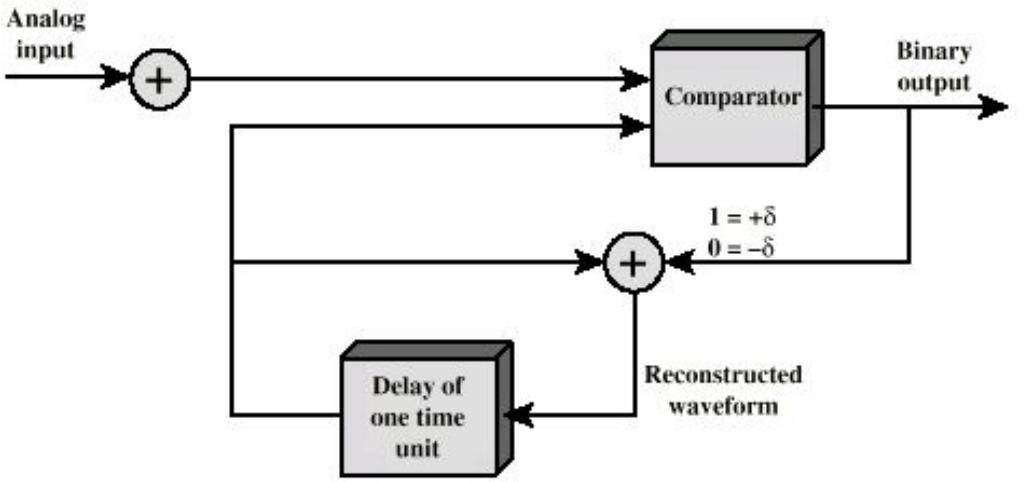
Delta modulation components



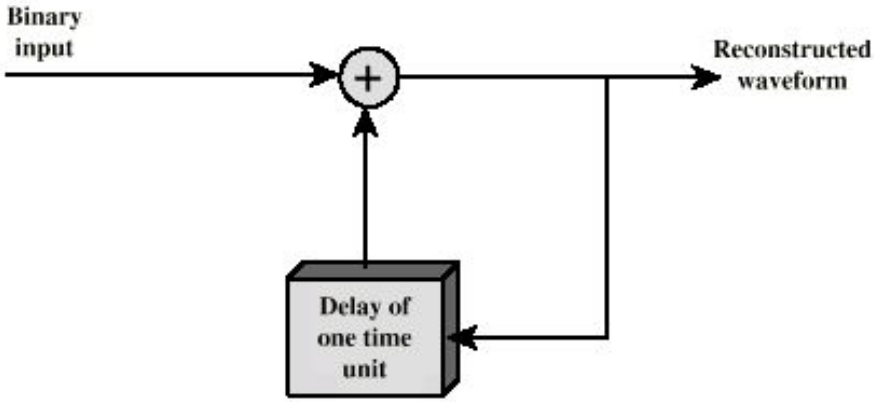
Delta demodulation components



Delta Modulation - Operation



(a) Transmission



(b) Reception

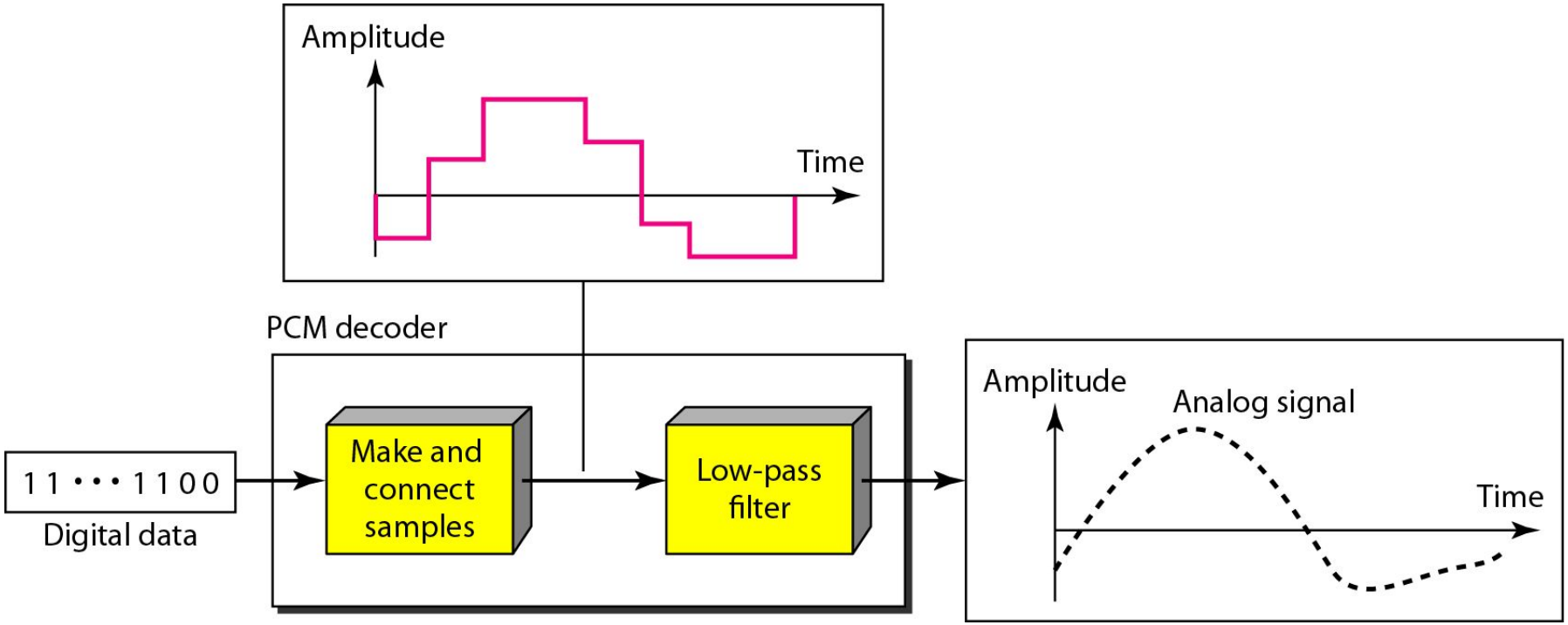
Delta PCM (DPCM)

- Instead of using one bit to indicate positive and negative differences, we can use more bits -> quantization of the difference.
- Each bit code is used to represent the value of the difference.
- The more bits the more levels -> the higher the accuracy.

PCM Decoder

- To recover an analog signal from a digitized signal we follow the following steps:
  - We use a hold circuit that holds the amplitude value of a pulse till the next pulse arrives.
  - We pass this signal through a low pass filter with a cutoff frequency that is equal to the highest frequency in the pre-sampled signal.
- The higher the value of L (levels), the less distorted a signal is recovered.

Components of a PCM decoder



Example

We have a low-pass analog signal of 4 kHz. If we send the analog signal, we need a channel with a minimum bandwidth of 4 kHz. If we digitize the signal and send 8 bits per sample, we need a channel with a minimum bandwidth of 8 × 4 kHz = 32 kHz.

Transmission modes

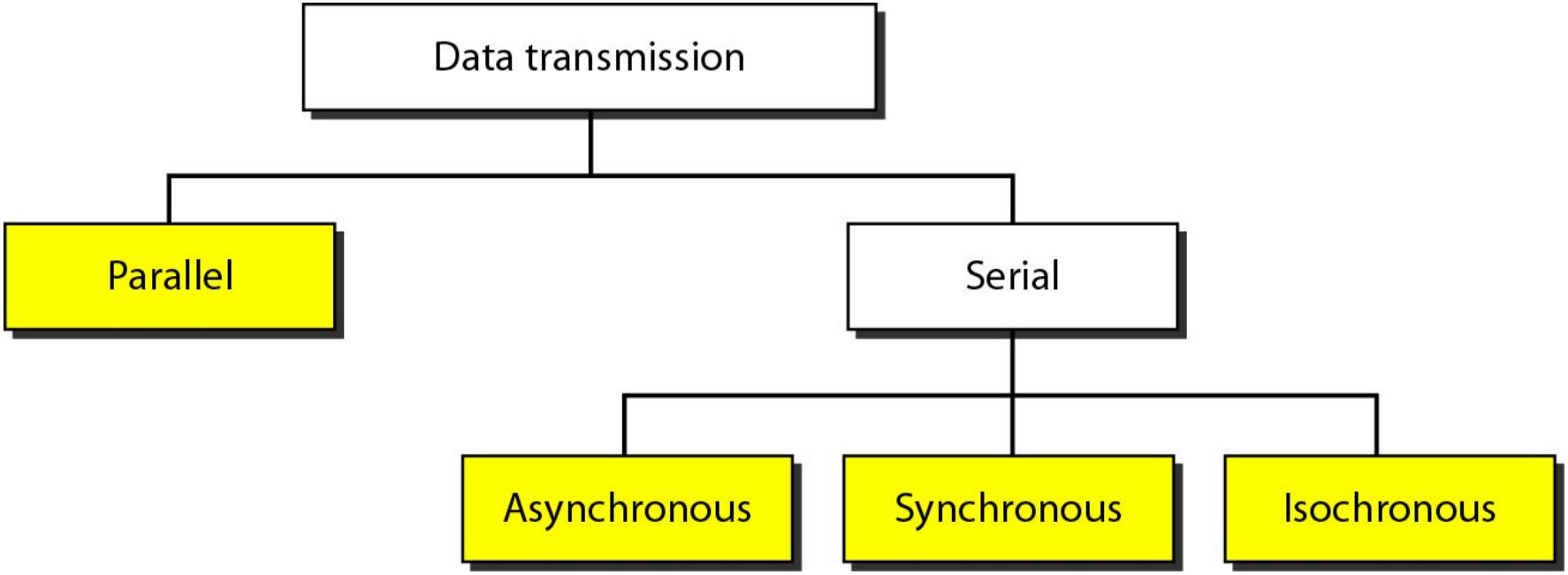
The transmission of binary data across a link can be accomplished in either parallel or serial mode.

- ***In parallel mode, multiple bits are sent with each clock tick.***
- ***In serial mode, 1 bit is sent with each clock tick.***

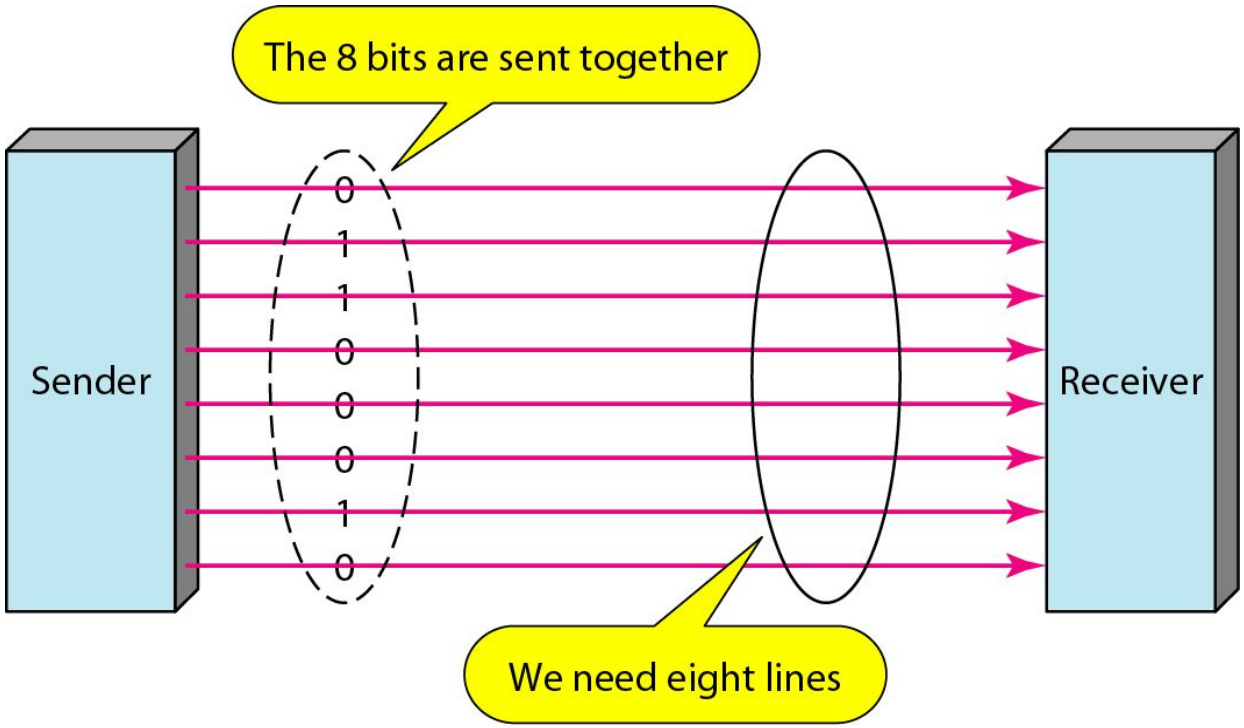
While there is only one way to send parallel data, there are three subclasses of serial transmission: asynchronous, synchronous, and isochronous.

- **Parallel Transmission**
- **Serial Transmission**

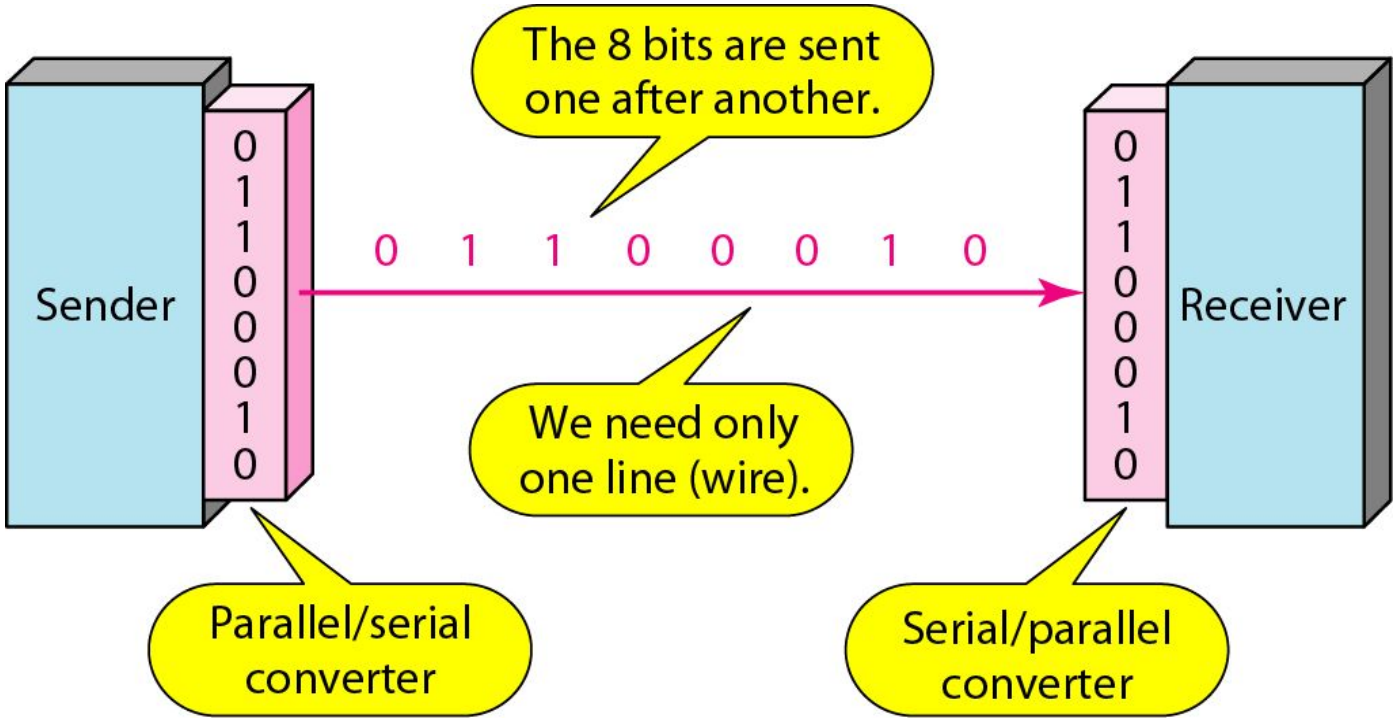
Data transmission and modes



Parallel transmission



Serial transmission



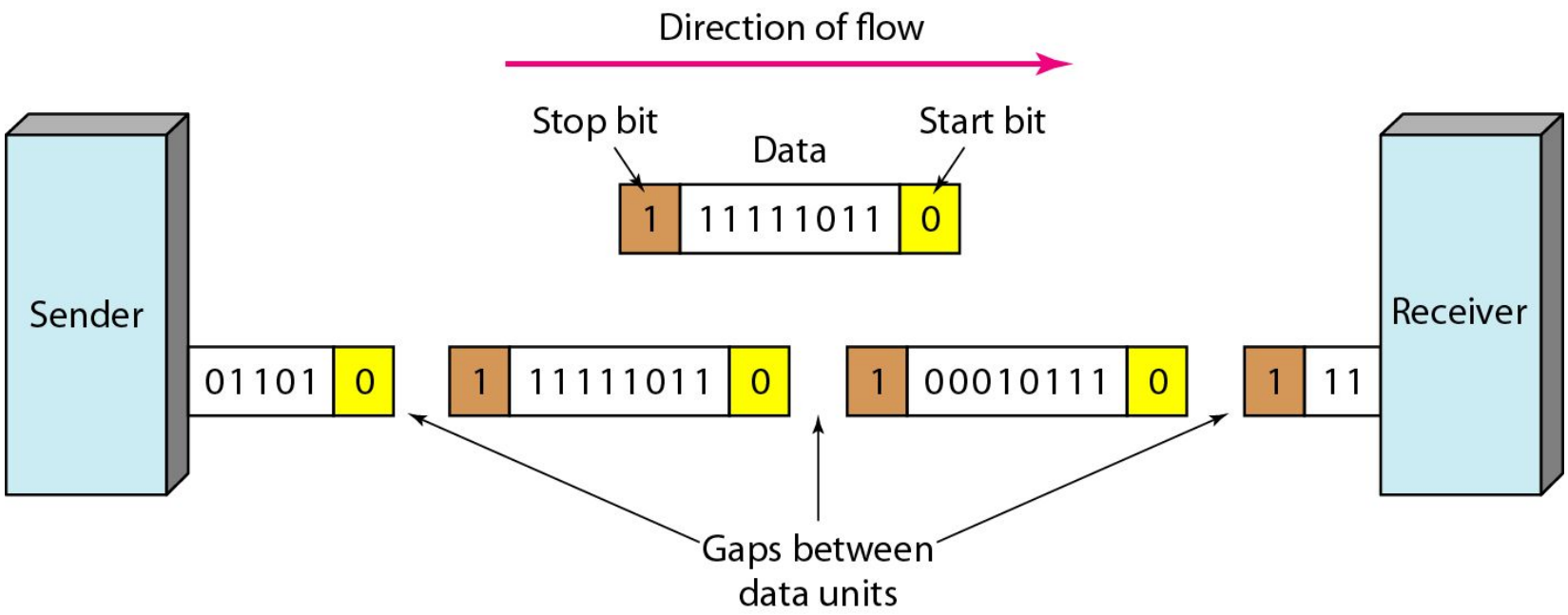
Note

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Note

Asynchronous here means "asynchronous at the byte level," but the bits are still synchronized; their durations are the same.

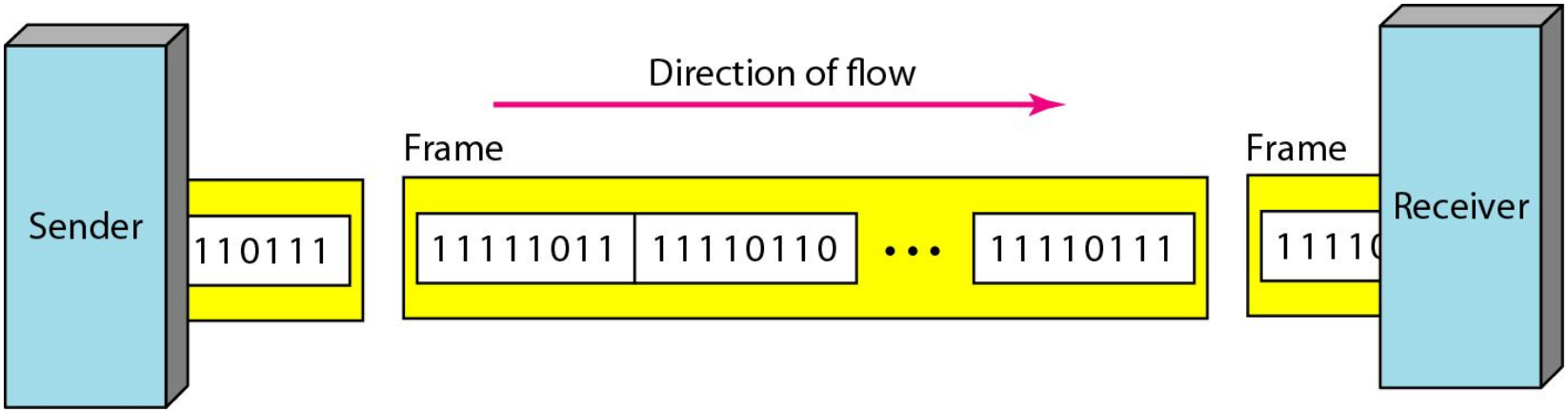
Asynchronous transmission



Note

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits. The bits are usually sent as bytes and many bytes are grouped in a frame. A frame is identified with a start and an end byte.

Synchronous transmission



Isochronous

- In isochronous transmission we cannot have uneven gaps between frames.
- Transmission of bits is fixed with equal gaps.